

```
from google.colab import files
uploaded = files.upload()
```

[Choose Files](#) groceries - groceries.csv
groceries - groceries.csv(text/csv) - 802870 bytes, last modified: 4/24/2026 - 100% done
 Saving groceries - groceries.csv to groceries - groceries (1).csv

```
import pandas as pd
from mlxtend.preprocessing import TransactionEncoder
from mlxtend.frequent_patterns import apriori, association_rules
import warnings
warnings.filterwarnings('ignore')
```

```
data = pd.read_csv("groceries - groceries.csv", header=None)
data.head()
```

	0	1	2	3	4	5	6	7	8	9	...
0	Item(s)	Item 1	Item 2	Item 3	Item 4	Item 5	Item 6	Item 7	Item 8	Item 9	...
1	4	citrus fruit	semi-finished bread	margarine	ready soups	NaN	NaN	NaN	NaN	NaN	...
2	3	tropical fruit	yogurt	coffee	NaN	NaN	NaN	NaN	NaN	NaN	...
3	1	whole milk	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...
4	4	pip fruit	yogurt	cream cheese	meat spreads	NaN	NaN	NaN	NaN	NaN	...
-	-	-	-	-	-	-	-	-	-	-	-

```
transactions = []

for i in range(len(data)):
    transactions.append(data.iloc[i,0].split(','))
```

```
te = TransactionEncoder()
te_array = te.fit(transactions).transform(transactions)

df = pd.DataFrame(te_array, columns=te.columns_)
df.head()
```

	1	10	11	12	13	14	15	16	17	18	...	29
0	False	False	False	False	False	False	False	False	False	False	...	False
1	False	False	False	False	False	False	False	False	False	False	...	False
2	False	False	False	False	False	False	False	False	False	False	...	False
3	True	False	False	False	False	False	False	False	False	False	...	False
4	False	False	False	False	False	False	False	False	False	False	...	False

5 rows × 30 columns

```
frequent_itemsets = apriori(df, min_support=0.01, use_colnames=True)
```

```
rules = association_rules(frequent_itemsets, metric="confidence", min_threshold=0.5)
```

```
from IPython.display import display
```

```
display(rules[['antecedents', 'consequents', 'support', 'confidence', 'lift']])
```

No entries						Filter		
index	antecedents	consequents	support	confidence	lift			

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